NEUE FISCHE · AI PROJECT MANAGEMENT

EDA Project

Exploratory Data Analysis · Real estate



TEAM MEMBERS

Trupti, Serena

PRESENTATION DATE

07.11.2026

OVERVIEW

Today’s agenda

01 **The client**
Who William is and what he needs

02 **How we worked**
From raw data to testable questions

03 **What we found**
Three hypotheses, and what each means for William

PART 01

Our client & modus operandi

01

Who William is and how we worked

01 — THE CLIENT

Two homes, two opposite priorities



William Rodriguez

Household of 2

The brief

2 people, buying two homes

Country house

Buy at the best timing · non-renovated

City house

Fast purchase · central location

02 — OUR APPROACH

Clean + explore took 65% of our time

01
Brief
Turn William's needs into testable questions

→

02
Clean
Missing values, duplicates, data types

→

03
Explore
New features: $/sqft, distance to centre, season

→

04
Test
Three hypotheses on filtered segments

→

05
Advise
Two concrete buying strategies

WHERE OUR TIME WENT

Data first, hypotheses second



03 — HYPOTHESES

Three questions we asked the data

H1

COUNTRY · TIMING

THE QUESTION

When should William buy a non-renovated country house?

OUR HYPOTHESIS

Non-renovated country houses (400–2,000 sqft) have a lower median price per sqft in Oct–Feb than in Mar–Sep.

H2

CITY · LOCATION

THE QUESTION

Where are compact city homes easiest to find?

OUR HYPOTHESIS

1–2 bedroom houses are more concentrated within 10 miles of the city centre than in outer areas, and cost more per sqft.

H3

COUNTRY · CONDITION

THE QUESTION

Does skipping the renovation save money?

OUR HYPOTHESIS

Unrenovated country houses sell at a 15–20% lower price per sqft than renovated homes in the same areas.

Every clue checked before the verdict

Survey the scene

We explored the dataset to understand how it was built.

It revealed some problems

Missing values

Found: 6,754 gaps in waterfront, view, yr_renovated, sqft_basement

Action: sqft_basement calculated (living − above); the rest filled with 0

Wrong formats

date: string → datetime

zipcode: int → string

id: int → text

Odd ones out

Found: 1 sale, a 33-bedroom house

Action: removed after a bedroom count check

PART 02

Hypothesis 1

Country house · Best timing

H1

H1 · COUNTRY HOUSE · TIMING — THE TEST

When should William buy a non-renovated country house?

OUR HYPOTHESIS

Non-renovated country houses (400–2,000 sqft) have a lower median price per sqft in Oct–Feb than in Mar–Sep.

01

Variables we used

yr_renovated, sqft_living, sqft_lot, lat, long, date, price

02

A new variable: country house

Defined by distance from downtown Seattle and lot size

03

Thresholds set by the data

Percentiles, not gut feeling: ≥ 16 mi (75th) and lot ≥ 20,000 sqft (90th)

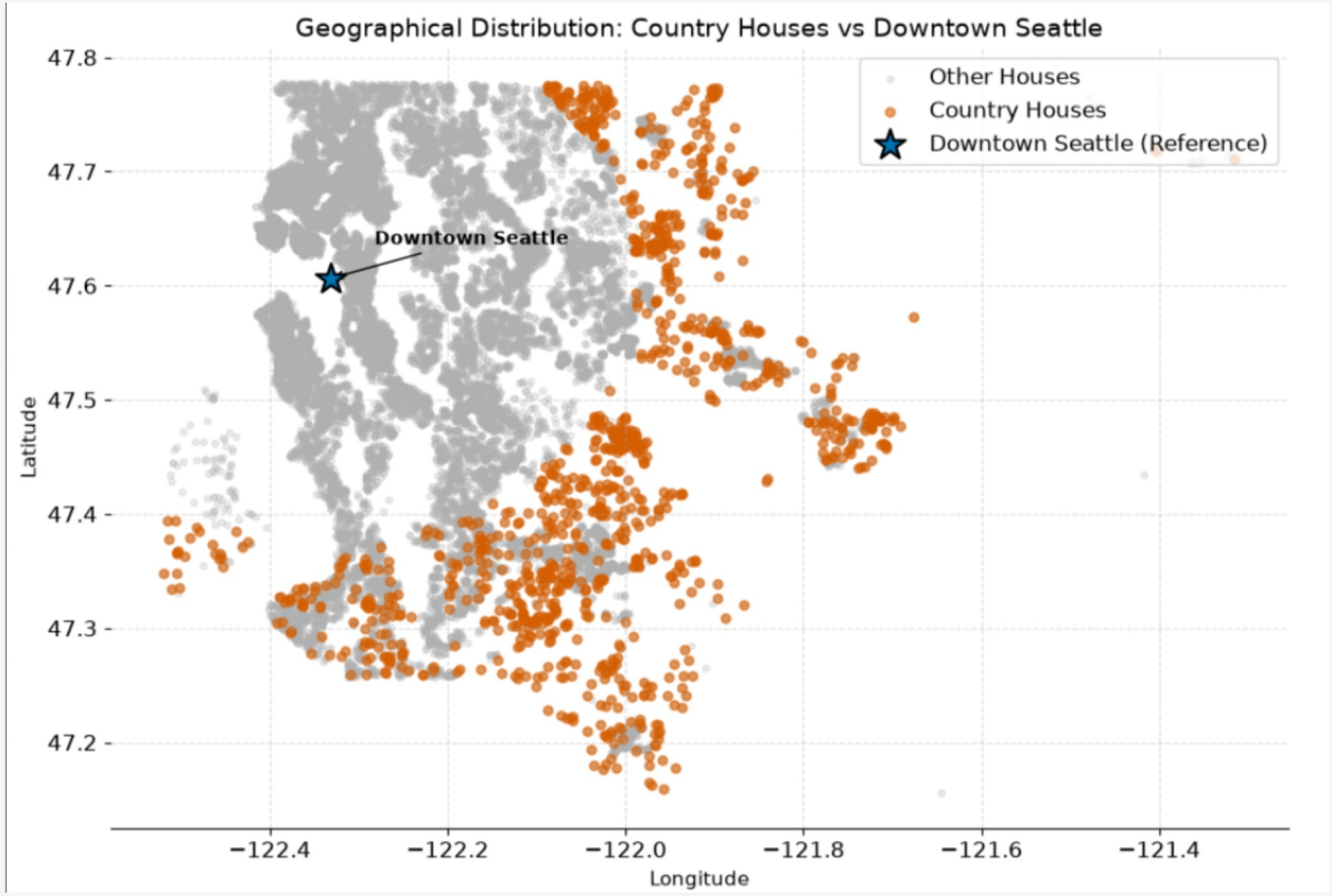
04

Suitable for two people

Kept only homes that fit a couple: 400–2,000 sqft · 1–2 bedrooms

H1 · COUNTRY HOUSE · WHERE THEY ARE

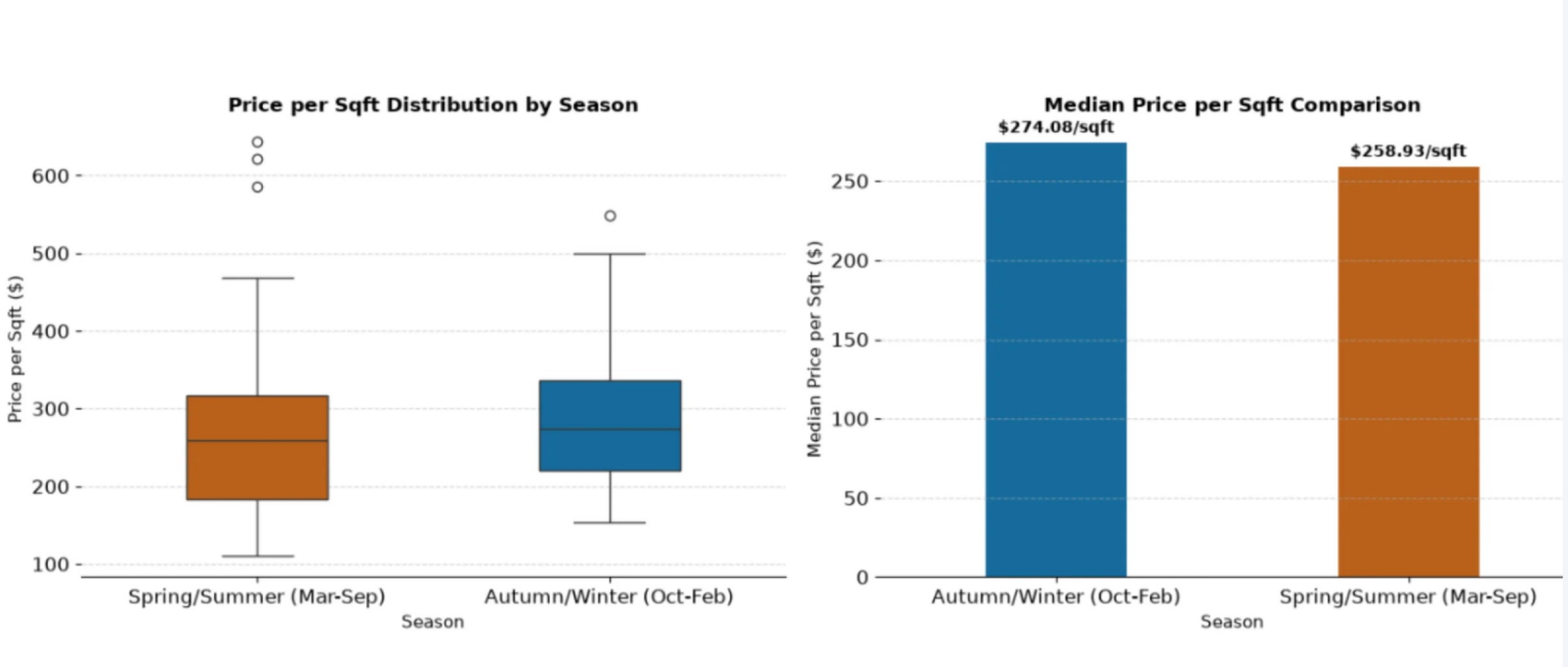
1,081 country houses ring the Seattle area





H1 · COUNTRY HOUSE · TIMING — THE VERDICT

Off-season buying does not save money



REJECTED

+5.9%

median $/sqft in Oct–Feb ($274) vs Mar–Sep ($259) · not significant, p = 0.26

→ FOR WILLIAM

No need to wait for winter: buy when the right country house comes up

No seasonal saving: 26 Oct–Feb vs 49 Mar–Sep sales

PART 03

Hypothesis 2

City house · Central location

H2

H2 · CITY HOUSE · LOCATION — THE TEST

Where are compact city homes easiest to find?

OUR HYPOTHESIS

1–2 bedroom compact houses are significantly more concentrated within 10 miles of the city centre than in outer areas, and command a higher price per sqft.

WHAT IT TOOK TO TEST IT

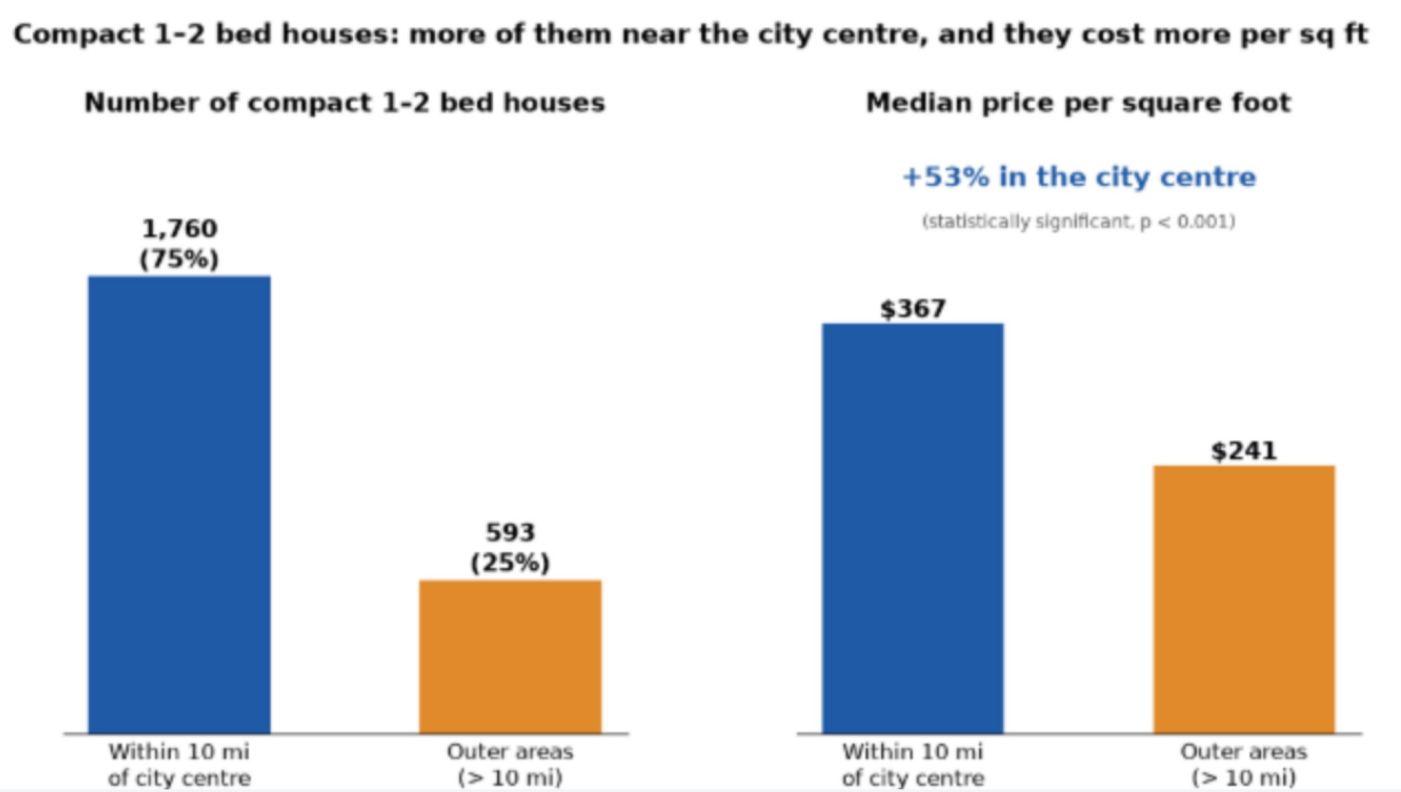
01 **Variables we used**
lat, long, bedrooms, sqft_living, price

02 **Who we compared**
1–2 bed, 400–1,500 sqft · within vs beyond 10 mi

03 **How we compared**
Share of homes and median $/sqft · Mann-Whitney U

H2 · CITY HOUSE · LOCATION — THE VERDICT

75% of compact homes sit within 10 miles



CONFIRMED

75%

of 1–2 bed homes within 10 mi (1,760 of 2,353) · +53% $/sqft vs outer areas ($367 vs $241)

→ FOR WILLIAM

Search within 10 miles of the centre to buy fast

Budget for a ~53% premium per sqft: ~$128,000 more on a 1,010 sqft home

PART 04

Hypothesis 3

Country house · Non-renovated

H3

H3 · COUNTRY HOUSE · CONDITION — THE TEST

Does skipping the renovation save money?

OUR HYPOTHESIS

Unrenovated country houses sell at a 15–20% lower price per sqft than renovated homes in the same country areas.

WHAT IT TOOK TO TEST IT

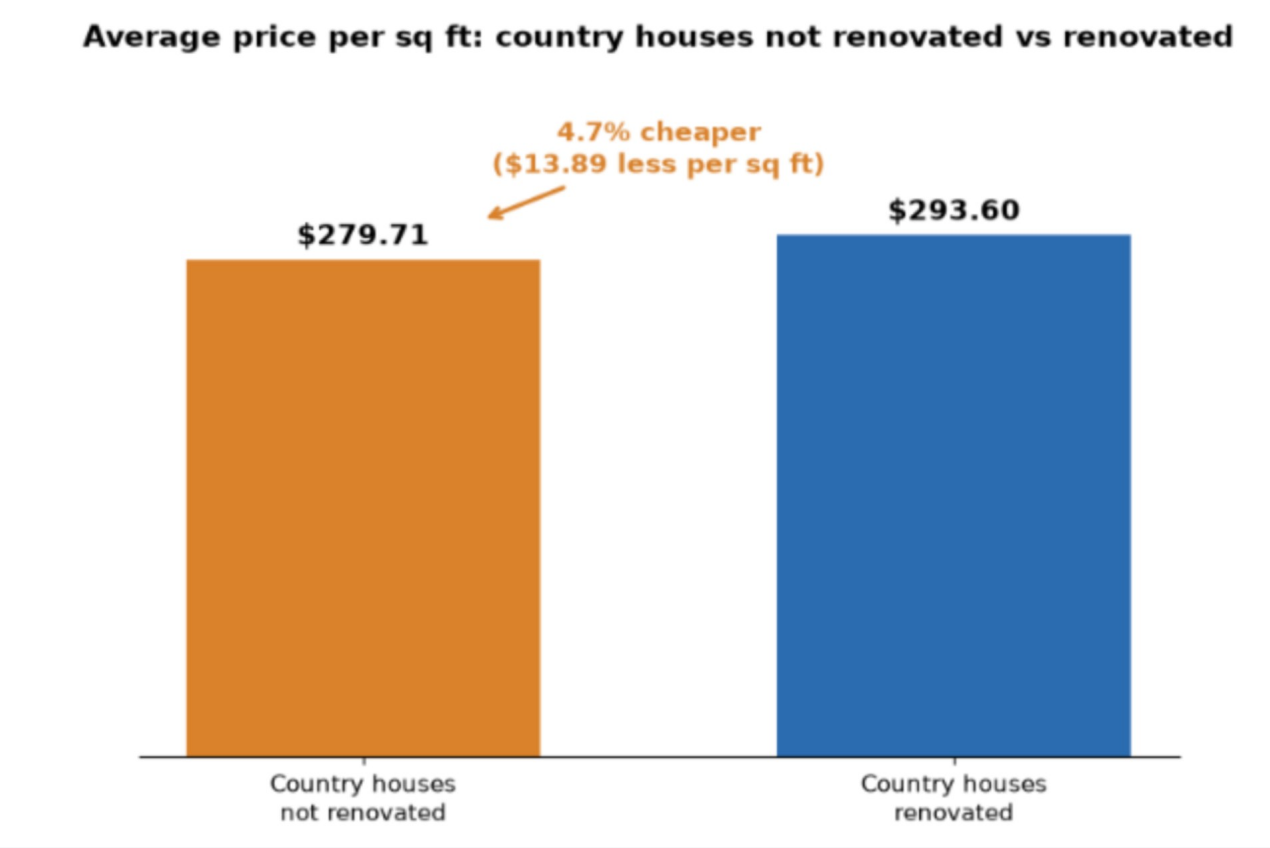
01 **Variables we used**
yr_renovated, sqft_living, sqft_lot, lat, long, price

02 **Who we compared**
Country houses · 75 never renovated vs 6 renovated

03 **How we compared**
Mean $/sqft gap vs the 15–20% target · α = 0.05

H3 · COUNTRY HOUSE · CONDITION — THE VERDICT

Non renovated saves only 4.7% per sqft



REJECTED

−4.7%

mean $/sqft: $279.71 unrenovated vs $293.60 renovated · target 15–20% · only 6 renovated homes

→ FOR WILLIAM

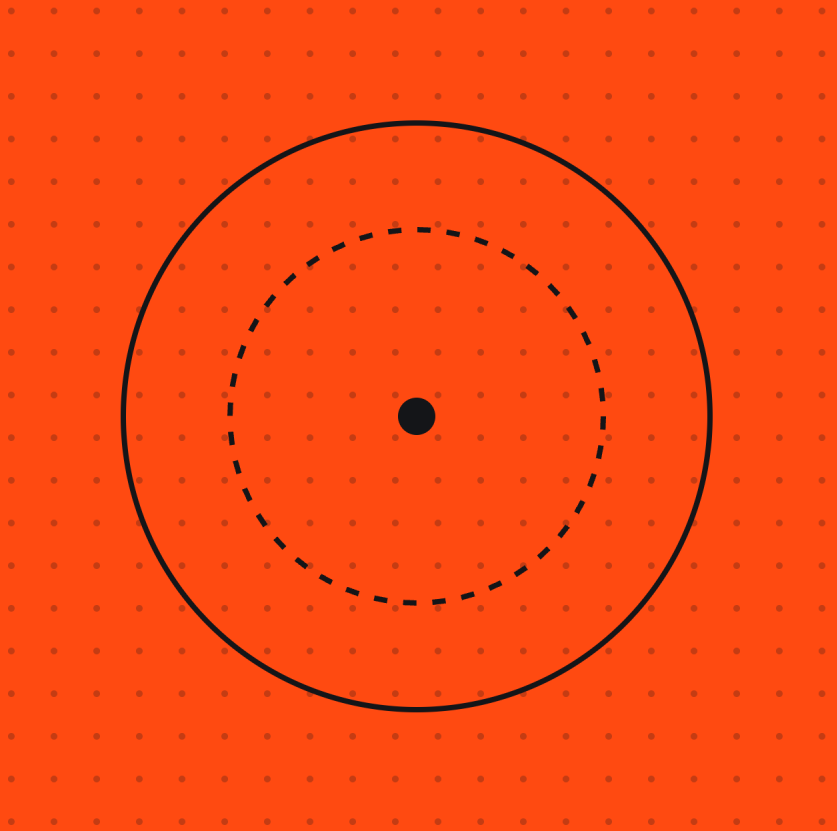
Don’t count on a big discount for skipping the renovation

Saving: ~$16,700 on a 1,200 sqft home

Q&A

Thank you.

Questions?



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